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MILITARY SPECIFICATION

INTRAVENOUS INJECTION SET

This limited coordination military specification has been prepared by the Defense Personnel Support Center based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-I-36996. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1. Scope. This specification covers sterile, disposable vented intravenous injection sets (infusion apparatus) designed for use in the administration of parenteral solutions from standard glass solution containers without airways.

1.2 Classification. Intravenous injection sets shall be of the following types as specified (see 6.2):

Type I. - Regular set (20 drops/ml.)

Type II. - Pediatric set (60 drops/ml.)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-N-196	Needles, Hypodermic, for Luer-Syringes.
MMM-A-250	Adhesive, Water Resistant, for Sealing Fiberboard Boxes
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Material, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

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Military

MIL-P-116	Preservation, Methods of
MIL-L-10547	Liners, Case and Sheet, Overwrap; Waterproof for Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Palletized Unit Loads.
MIL-STD-794	Part & Equipment, Procedures for Packaging and Packing of

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained as directed by the contracting officer).

2.2. Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposals shall apply:

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE
Bureau of Biologics. Food and Drug Administration
Code of Federal Regulation part 273. Minimum Requirements (as applicable).

(Application for copies should be addressed to the Food and Drug Administration, 5600 Fishers Lane, Rockville, Maryland 20852.)

AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

ANSI/HIMA MD70.1 Luer Taper Fittings for Medical Materiel.

(Application for copies should be addressed to the Publication Sales Dept. STD, American Institute of Physics, 335 E. 45th St., NY, NY 10017).

U. S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States (USP)

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material.

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3.1.1. Rubber and plastic components. All rubber and plastic components shall be of the best quality and suitable in all respects for the intended purpose. All rubber and plastic components shall be sterilizable, free from pyrogens, compatible, and safe when tested as specified in section 4. Those rubber and plastic components coming in contact with solutions being administered, and end protectors, shall contain not more than 5 parts per million (p.p.m.) total of lead, barium, and cadmium when tested as specified in 4.4.1, or 4.4.1.1. The plastics shall meet the requirements of the Pharmacopeia of the United States (USP) for Class IV plastics and for transfusion and infusion assemblies when tested as specified in 4.4.2.

3.1.1.1 Plastic tubing. Plastic tubing shall be clear, colorless, transparent, flexible polyvinyl chloride, having a hardness of 68 + 6, -5 Shore A durometer, when tested as specified in 4.4.3. The tubing shall be flexible and pliable, but sufficiently rigid to prevent collapse during normal use. The tubing shall be clean and free of particles and shall be free from all liquid substances which oxidize, exude, or sweat-out to the surface of the tubing.

3.1.2 Needles.

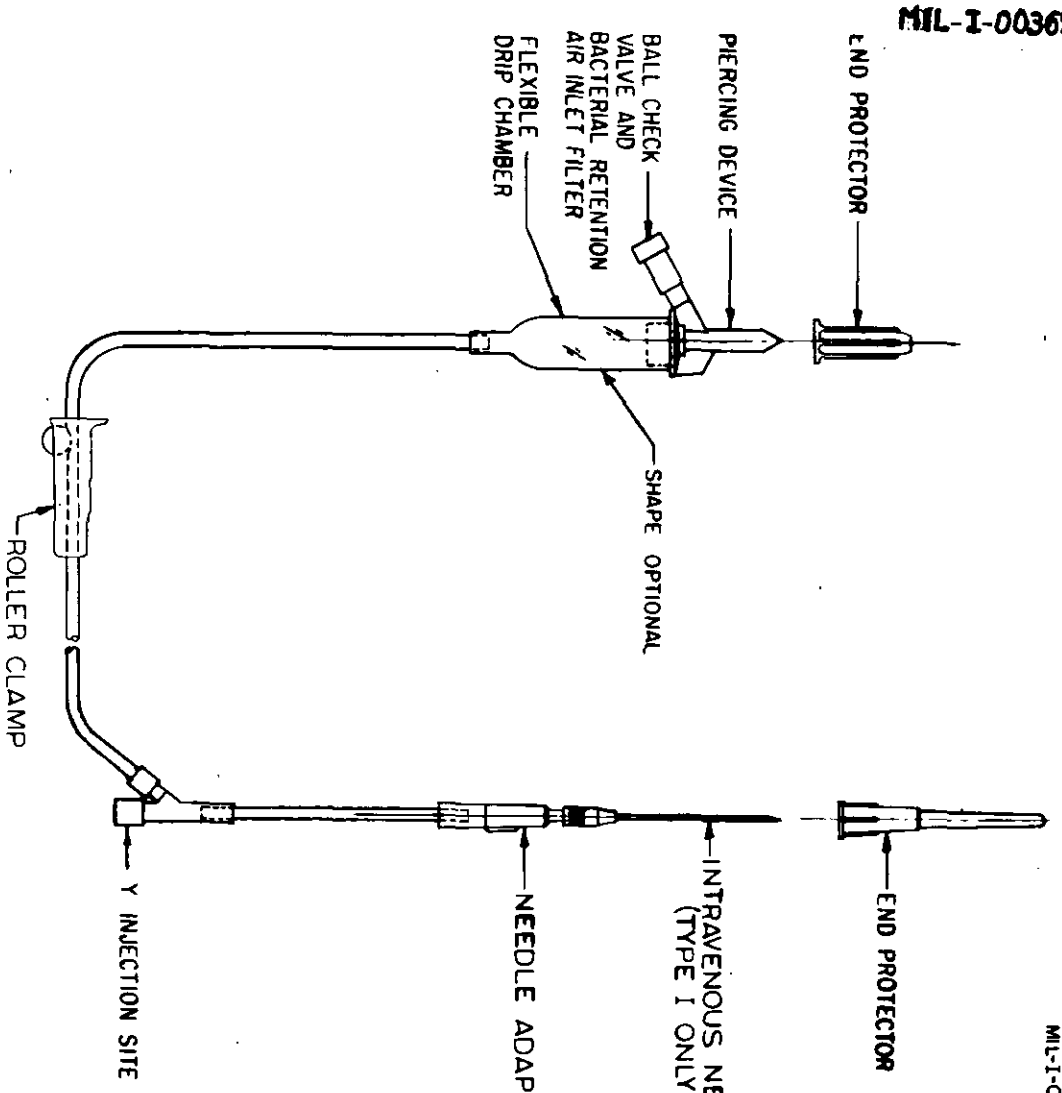
3.1.2.1 Cannulae of needles. Cannulae of needles shall be fabricated from 304 AISI stainless steel, and shall meet the corrosion-resistance, elasticity, and resistance-to-breakage tests as specified in GG-N-196. The cannulae shall be fabricated from either seamless drawn, or welded and drawn tubing.

3.1.2.2 Hub. Hubs shall be fabricated from a suitable plastic (containing no exudable toxic materials), aluminum alloy, nickel or chromium-nickel plated brass. Aluminum hubs shall be degreased and then either anodized, or chemically passivated.

3.1.3 Other material. All other material hereinafter specified shall be of the best quality and suitable in all respects for the purpose intended.

3.2 Type I - Regular set. The type I set shall be a sterile, non-pyrogenic, safe (nontoxic), disposable intravenous injection set (infusion device) for use in the administration of parenteral solutions from standard glass solution containers (see 3.2.2). The style and design shall be in accordance with Figure 1 and as specified herein. The set shall have a piercing device, a flexible plastic calibrated drip chamber, an air vent with a check valve covered by a bacterial filter, one female Luer adapter, a length of plastic tubing, an injection site, a male Luer adapter, an intravenous needle, a roller-type flow control device, and end protectors. The drip chamber shall be molded of suitable, flexible, colorless clear plastic. The piercing device shall be permanently attached to the drip chamber in a leakproof manner. The assembled drip chamber shall be so shaped as to provide a convenient hand grip for the insertion of the piercing device into the stopper or port of the solution container. The assembly of the set shall be structurally sound and shall not leak air or fluid during use. Open ends of the set shall be covered (sealed) with readily removable covers, or end protectors, which shall maintain the sterility and freedom from pyrogens of the fluid path after sterilization.

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INTRAVENOUS INJECTION SET

FIGURE 1

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3.2.1 Drip chamber. The drip chamber housing shall have an overall length of about 2-7/8 inches, measured from the top of the open drip barrel to the bottom of the tubing connector end. The inside diameter at the bottom stem (where plastic tubing connects) shall be 1/16 to 9/64 inch. The volume capacity of the drip chamber measured up to the level of the drip orifice shall be 3 to 7 ml.

3.2.2 Cap (piercing device). The cap shall consist of an integrally molded unit including a piercing device (spike), and an air inlet with built in check valve. The piercing device shall be a suitable rigid plastic, dual-channeled, conical spike, one channel of which shall serve as an air inlet, the other channel being extended to form the drip orifice. The piercing device tip shall be 7/8 to one inch in length, with a diameter of 0.258 to 0.267 inch at the base. The piercing device tip shall be sharpened to such a point as to be suitable for puncture of, and insertion into, the stopper or port of a standard solution container (as specified herein) without undue pressure (see 3.2.2.1). Edges of fluid and air inlet passage openings shall be rounded or dulled to prevent stopper coring. The piercing device shall be ruggedly designed to prevent coring or shearing, and shall be structurally sound. It shall be firmly retained in a leakproof manner when inserted in the stopper or port of a standard solution container. The piercing device shall not break or leak air or fluid during use. It shall not be brittle or soft, or become brittle or soft under normal storage conditions. The standard solution container stopper referenced above shall be in accordance with requirements contained in the Defense Medical Purchase Description for NSN 6505-00-975-0604, Dextrose Injection, USP. The passages in the piercing device shall be properly designed to prevent air from passing directly from the air passage to the fluid passage under maximum flow conditions. The air inlet passage shall contain a valve seat and a stainless steel check ball and vent the solution container through the bacterial filter during administration of solution. The air inlet shall not leak when tested as specified in 4.4.8. This air inlet opening shall be capped with a removable and replaceable porous air filter which shall prevent the ingress of bacteria (see 3.1.2.2), and which shall serve as an air filter during storage and use. The filter cap shall form a snug fit with, and shall be retained on, the opening until manually removed. The fluid passage in the spike shall extend into the drip chamber, and shall be calibrated to deliver 15-20 drops per milliliter (ml.) when tested as specified in 4.4.5. Shall maintain the 15-20 drops per ml drop size during administration of intravenous solutions for a minimum of 48 hours.
The instructions supplied with the set (see 3.10) shall include the drop size (i.e.), the number of drops per ml.). The drop size of each set shall not vary by more than plus or minus 3 drops per ml from the drop size indicated in the instructions.

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3.2.2.1 Durability of piercing device. When tested as specified in 4.4.6 the piercing device, or the cap, shall not loosen or permit leakage, nor shall the device or cap become damaged. The piercing device shall be capable of being inserted into the standard solution containers, as described in 4.4.6, without difficulty, the tip (or point) of the device not breaking or bending to the degree of causing difficulty in insertion into standard solution containers.

3.2.2.2 Air inletting bacterial filter. The air inletting bacterial filter shall be of such design and material as to comply with the following performance requirements when tested as specified in 4.4.7:

(a) Bacterial retention. The filter shall be such that the rate of contamination of the test media shall be not greater than 0.1% (1-1000).

(b) Air passage. The filter shall pass air at a minimum rate of 150 ml. per minute.

(c) Resistance to water. The filter shall support a 30-inch static head of water for 45 seconds with no wetting of the filter element.

3.2.3 Plastic tubing. The plastic tubing shall be fabricated of material as specified in 3.1.1.1 and shall be a minimum of 60 inches. It shall have an ID of .095 to .105 and an OD of .130 to .145 inch. It shall be firmly and permanently secured, in a leakproof manner, to the drip chamber at one end and to a male needle adapter at the other end in such a manner as to form an airtight, leakproof connection which shall not separate when tested as specified in 4.4.8. A "Y" type injection site fabricated of clear or translucent rigid plastic shall be connected to the tubing at a nominal distance of six inches above the needle adapter. Two of the legs shall be in line, and the third leg at a nominal 30° angle. Top leg of inline portion of "Y" shall have a resealable diaphragm or plug securely fitted. Tubing is fastened to other two legs with single leg toward the needle adapter.

3.2.4 Rubber diaphragm. The rubber diaphragm or plug on the medicinal entry assembly leg shall be molded from natural, non-oxidizing, low-sulfur, wax-free rubber of 45 $\pm$ 5 Shore A Durometer. The diaphragm or plug shall be so secured that after puncture and withdrawal of the needle in accordance with 4.4.9 there shall be no separation or leakage.

3.2.5 Control clamp. Control clamp shall be fabricated of plastic, and be a roller type clamp from which the roller cannot be accidentally removed. The clamp shall be capable of withstanding a 3.65 kilogram load when tested as specified in 4.4.10.1 without the roller separating from the body of the clamp. The clamp shall be supplied on tubing in the open position and shall provide a constant setting and quick shut off from the operating position, shall be movable, and shall be capable of one-handed operation. To assure constant delivery of fluid, it shall be impossible for the clamp setting to be changed drastically by an accidental jerk on the tubing. When tested as specified in 4.4.10, the clamp shall be capable of accomplishing complete quick shutoff (i.e. no falling drop in drip chamber or from needle adapter (3.2.6), shall regulate flow of the solution from complete shutoff to maximum flow, and with a

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constant head of water shall maintain a constant flow rate at any point between complete shutoff and maximum flow. In addition, when the clamp is operated from its fully open position to complete shutoff, and this operation is repeated ten times while the clamp is in the same position on the tubing, the tubing shall not be cut, pierced, or otherwise damaged so as to cause leakage or restriction of flow of solution.

3.2.6 Male needle adapter. The male needle adapter shall be fabricated of suitable plastic and shall be tapered to fit a needle having a standard Luer hub in accordance with ANSI/HIMA Standard MD70.1 in a tight leakproof manner (see 3.8). The needle adapter shall not separate from the tubing or show evidence of leakage when tested as specified in 4.4.8.

3.2.7 Intravenous needle. The needle shall be a 19 or 20 gage, 1-1/2 inch cannula intravenous short bevel needle. Except as specified herein, the needle shall be in accordance with requirements of GG-N-196 for type I, point style b, hub style 1 or 2. The cannula and hub shall be fabricated of material as specified in 3.1.2.1 and 3.1.2.2, respectively. The hub cavity shall be constructed in accordance with ANSI/HIMA Standard MD70.1. The needle shall fit the needle adapter in such a manner as to make a tight, leakproof connection. The hub cavity shall be tested for diameter and depth and fluid tightness as specified in 4.4.11. Hubs shall fit the plug gage within the "GO" - "NO GO" limits of the ANSI/HIMA Standard MD70.1. Hubs which do not fit the plug gage within these limits shall be considered not acceptable. In addition, when the hub is fitted onto the plug gage as specified in 4.4.11.1, there shall be no movement of the hub on the gage when the cannula is pressed lightly from side to side in all directions. When tested for fluid tightness as specified in 4.4.11.2 the swaged connection of the cannula with the hub, and the connection of the hub with the test gage, shall not show leakage exceeding the formation of a falling drop. The needle shall be supplied with hub attached to the tapered end of the needle adapter, or in close approximation for such attachment. The needle shall be covered by the end protector in such a manner as to be readily and easily attached to the needle adapter and retain its sterility. The needle marking required by GG-N-196 may be omitted.

3.2.8 End protector. The protector shall be attached to the needle adapter while the needle is contained within the protector in a manner which will prevent contact of the needle point with the inside end of the protector and also provide a loose fit over the Luer end of the needle adapter. End protectors shall be readily removable, plastic and constructed in such a manner that when placed over each open end of the intravenous injection set it shall, after sterilization, maintain sterility, freedom from pyrogens, immobility of the needle, and integrity of the entire fluid path of the infusion device until manually removed.

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3.3. Type II - Pediatric set. Type II pediatric intravenous injection set shall comply with the requirements specified herein for the type I (Regular) set (see 3.2) except that the fluid passage in the spike shall be calibrated to deliver 60 ± 9 drops per milliliter when tested as specified in 4.4.5, and shall be supplied without a needle. Shall maintain the 60 drop per milliliter drop size during administration of intravenous solutions for a minimum of 48 hours at approximately 60 drops per minute. The plastic tubing shall be as specified in 3.2.3 except that it shall be a minimum of 70 inches in length. The set assembly shall be structurally sound and shall not leak air or fluid during use.

3.4 Manufacturer's record. The manufacturer shall place a lot control number on the paper label, or on the printed carton, which can be verified against the manufacturer's records of all tests made available for inspection by any duly qualified medical officer of the Armed Forces or any official of the Federal Food and Drug Administration, or any other duly authorized Government representative. No shipment shall be made until all the tests have been completed, verified, and recorded in the permanent record of the manufacturer.

3.5 Sterilization, safety, and freedom from pyrogens. All equipment shall be safe. Fluid paths, end connectors, and needles of type I sets, and fluid paths and end connectors of type II sets shall be sterile and pyrogen-free. The fluid path, end connectors, and needles shall be pyrogen-free when tested as specified in 4.4.12 and sterile when tested as specified in 4.4.13. The sets shall be safe when tested for safety as specified in 4.4.14.

3.6 Certificates. Certificates of sterility, freedom from pyrogens, safety, and bacterial retention as specified below, shall be submitted. For destination inspection, certificates shall be submitted by the contractor directly to Directorate of Medical Materiel, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA 19101, ATTN: Quality Assurance Branch, and one copy of each shall be forwarded with each shipment to the consignee. For source inspection certificates shall be furnished to the procuring activity via the government inspector.

3.6.1 Certificate of sterility. A certificate of sterility, listing all sterility lot or control numbers, shall state that samples from each sterility lot or control have been tested for sterility, as required in 3.5 and were found to be sterile. The certificate shall reference the method used in determining sterility.

3.6.2 Certificate of freedom from pyrogens. A certificate of freedom from pyrogens, listing all lot or control numbers, shall state that samples from each lot or control have been tested for pyrogens as specified in 3.5 and were found to comply with such requirements. The certificate shall reference the method used in determining freedom from pyrogens.

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3.6.3 Certificate of safety. A certificate of safety, listing all lot or control numbers, shall state that samples from each lot or control have been tested for safety as specified in 3.5, and were found to comply with such requirements. The certificate shall reference the method used in determining safety.

3.7 Compatibility. The materials of which the type I and type II sets are manufactured shall not react with, or be acted upon by, the liquids with which they will be in contact during normal use and operation, nor shall there by any materials incorporated in the set which will cause any deleterious effect when the set is used for the purpose intended.

3.8 Leakage and separation of set. Type I and type II sets shall be tested for leakage and separation of set as specified in 4.4.8, and 4.4.8.1. Sets which show leakage or separation at any part of the set shall be rejected. A leak is defined as the formation of an air (gas) bubble or liquid as appropriate, which breaks away from the set within the specified test time.

3.9 Solution flow rate. When tested as specified in 4.4.15, the intravenous injection set, with the filter in its normal dry condition shall permit the passage of 100 cc of fluid in not more than 75 seconds (for Type I) or 5-1/2 minutes (for Type II), and not more than 120 seconds for (Type I) or 8-1/2 minutes (for Type II) with the filter wet (immerse filter in distilled water at room temperature for at least 10 minutes).

3.10 Instructions. Individual set carton shall have detailed instructions for proper use of the set for single use. Detailed instruction sheet for tandem use shall be inserted in each unit package (48's).

3.11 Workmanship. Workmanship shall be first class throughout. The intravenous injection sets shall be free from defects which detract from their appearance or impair their serviceability.

3.12 Government furnished material. The Government shall furnish the contractor the dextrose injection bags (NSN 6505-00-083-6538) needed to perform the leakage and drip test specified in 4.4.8.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the components of materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 Lot

4.2.1.1 Plastic. For purposes of this specification, a lot or batch shall consist of a quantity of plastic moldings or tubing of uniform composition produced under like conditions.

4.2.2 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in Table I. The unit of product for examination shall be one intravenous injection set.

Table I. Sampling for Examination

Type of defect	Inspection level	AQL (percent defective)
Major	S-4	1.0
Minor	S-4	2.5

4.2.3 For tests. Sampling for tests shall be in accordance with MIL-STD-105 and as indicated in Table II. The unit of product shall be one IV injection set.

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Table II. Sampling for Other Tests

Characteristic	Req.	Test procedure	Insp. level	AQL (percent defective)
Heavy metal ** (extractable)	3.1.1	4.4.1	S-1	*
Class IV Plastic **	3.1.1	4.4.2	S-2	*
Tubing hardness	3.1.1.1	4.4.3	S-2	*
Drop test	3.2.2 and 3.3	4.4.5	S-3	*
Piercing device durability	3.2.2.1	4.4.6	S-3	*
Air inletting bacterial filter	3.2.2.2	4.4.7.1	See 4.4.7.1	*
		4.4.7.2	S-2	*
		4.4.7.3	S-2	*
Leakage, and separation of set	3.2.2 3.2.6 and 3.8	4.4.8 and 4.4.8.1	S-3	*
Reseal (rubber diaphragm)	3.2.4	4.4.9	S-3	*
Control clamp	3.2.5	4.4.10 & 4.4.10.1	S-3	*
Intravenous needle	3.2.7	4.4.11, 4.4.11.1 & 4.4.11.2	S-3	*
Freedom from pyrogen	3.5	4.4.12	S-2	*
Sterility	3.5	4.4.13	See 4.4.13	*
Safety	3.5	4.4.14	S-2	*
Solution flow rate	3.9	4.4.15	S-3	*

* Acceptance number shall be zero for all samples sizes.

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** Each lot of product need not be tested provided evidence is furnished that samples of the formulation and identical method of processing had previously been tested, that they comply with USP requirements for Class IV plastics and heavy metals and that they are the same as being offered on this contract.

4.2.5 Sample unit usage. Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.2.6 For bacterial retention test. This test shall be performed at least once for each contract. The samples selected shall represente the exact composition, treatment, and construction to be used throughout the entire contract. Any change in the filter shall require additional testing. The number of samples specified in 4.4.7.1 shall be randomly selected from production lots of filters used. Failure of any sample to comply shall be cause for rejection of the filters represented.

4.3 Examination. The sets shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted in accordance with MIL-STD-105 and the classification of defects specified in Table III. Examination, however, shall not be restricted to the classified possible defects listed in Table III.

Table III. Classification of Defects

Categories	Defects
Critical	None defined.
Major	
101	End protectors or other components missing or dislodged.
102	End protectors not properly designed to permit flow of sterilizing gases, if used.
103	Not free of bloom or tackiness.
104	End protectors not properly designed to maintain sterility.
105	Style and design not as specified.
106	Presence of foreign matter.
107	Set not free of holes or cuts.
108	Dimensions not as specified.
Minor	
201	Packaging, packing and marking not as specified.
202	Instructions missing, incomplete or illegible.
203	Contractor's name or registered trademark, missing, incomplete, or illegible.

4.4. Tests.

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4.4.1 Heavy metals extractable test. A 2-gram composite sample shall be obtained from the rubber and plastic components. The material shall be cut into small slivers, so that no thickness exceeds 3 mm., and eluted for 72 hours at 50° C., 24 hours at 70° C., 60 minutes at 121° C. in one menstruum of 0.9% sodium chloride. Each gram of the material shall be covered with 5 milliliters (ml.) of the menstruum. The menstrua are collected, agitated after elution, and then submitted to spectographic atomic absorption, spectrophotometric or specific wet analysis for lead, barium, and cadmium. The metals are reported in parts per million. Thus 1.0 ppm in elute is equivalent to extracting 1.0 mg. from 200 g of test material dispersed in 1,000 ml. of elution menstruum.

4.4.1.1 Alternate heavy metals extractable test. Accurately weigh a 5-gram sample for each of the metals (barium, cadmium or lead) into a 1-oz. Vycor crucible. Reduce to ash in a muffle furnace at 650° C for eight hours. Let cool to room temperature and add 15 ml. of 1:1 hydrochloric acid. Bring to a boil on a hot plate, allow to boil for 2 to 3 minute, remove, let cool, and filter through Whatman #2 filter paper into a 50-ml. volumetric flask. Rinse the residue on the filter with deionised water and dilute to 50 ml. Prepare standard solutions of 1, 2, and 5 ppm for each of the metals, namely barium, cadmium, and lead. Determine the absorbances of these solutions at appropriate wave lengths using a Hollow Cathode Ba, Cd, or Pb lamp for each of the respective metals and a 10X scale expansion on a spectrophotometer, such as the Perkin-Elmer Model 303, Atomic Absorption Spectrophotometer. Determine the absorbance of the sample solutions using the same conditions as with the standards.

Calculation:

$$\frac{A \times F \times \text{dilution}}{W} = \text{ppm barium, cadmium, or lead}$$

A = Absorbance

F = Factor = $\left(\frac{\text{Concentration standard in ppm Ba, Cd, or PB}}{\text{absorbance}} \right)$*

W = Sample weight in grams

*The three standards are average to give ppm/unit A.

4.4.2 Biological tests. The plastics shall be tested in accordance with the USP for Class IV plastics. Sampling shall be as shown in Table II.

4.4.3 Tubing hardness. Plastic tubing hardness shall be determined in accordance with method 3021 of Federal Test Method Standard No. 601. The test shall be conducted at a room temperature of 27° ± 5° C (80° ± 90° F) on samples which had been conditioned at this temperature for not less than 3 hours prior to testing. The test sample shall be laminated to a thickness of not less than 1/4 inch. The gage reading shall be made 15 seconds after application of the instrument.

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4.4.5 Drop test. The intravenous injection set shall be tested at room temperature, using water at a rate of 100 ± 5 drops per minute. The number of drops required to deliver 2 ml. of water shall be counted, and the drops per ml. calculated from this value.

4.4.6 Piercing device durability test. Samples of finished sets, after sterilization, shall be used for this test. The piercing device (on assembled set) shall be inserted into, and removed from, a standard solution container (specified in 3.2.2) in the normal manner. This operation shall be repeated twice. The piercing device shall then be immersed in a solution of 5 percent dextrose in isotonic sodium chloride solution (in accordance with requirements of the USP monograph for "Dextrose and Sodium Chloride Injections") for 4 hours. The piercing device shall then be removed from the solution and immediately inserted into a standard solution container as specified in 3.2.2.

4.4.7 Air inletting bacterial filter.

4.4.7.1 Bacterial retention. The following test shall be performed on not less than 50 sample filters randomly selected to represent specific materials used in production batches of filters, and four additional test bottles shall be used for control test for each test period: Detach drip chamber assembly from each sample sterile intravenous injection set by sealing and cutting the plastic tube within 2 inches of drip chamber so that the tubing interior remains sterile. Regular parenteral 1000 ml. (1 liter) bottles having a standard closure shall be filled with 50 ml of fluid thioglycollate medium, USP, evacuated to at least 25 inches of mercury (Hg), sealed and sterilized. Introduce a sufficient concentration of A. aerogenes into a number 44 De Vilbiss pocket nebulizer, or similar apparatus. Connect nebulizer to air filter on a drip chamber assembly by means of gum rubber tubing with necessary reducers and connectors. Adjust the concentration of organisms in the nebulizer to obtain a minimum of 300 organisms in the air volume replacing the vacuum in the liter test bottle. Insert the piercing device of each prepared sample drip chamber into a bottle closure in the normal manner. When the vacuum in the liter test bottle has been displaced, disconnect the testing apparatus from the filter and shake the bottle thoroughly to mix the aerosol content and to wet all surfaces. Retain the drip chamber assembly in the bottle for venting and incubate at $31^{\circ} \pm 1^{\circ}$ C for seven days. As a control to assure that the aerosol is challenging the filter correctly, remove the air filter from the drip chamber assembly of two additional bottles at the start and two additional bottles at the end of each and every test period and connect the nebulizer. When the vacuum has been displaced, disconnect the nebulizer and replace the filter. After mixing the bottle contents, determine the bacterial count of the bottle. The rate of contamination shall be no greater than specified in 3.2.2.2.

4.4.7.2 Air passage. The air-inletting bacterial filter shall pass air as specified in 3.2.2.2, when tested assembled in the filter cap and under a pressure equal to a column of 16 inches of water. The flow shall be measured with a Brooks Rotameter, or equivalent apparatus.

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4.4.7.3 Resistance to water. The air-inletting bacterial filter shall be tested for resistance to water in its filter cap by placing the filter cap on a fitting connected to a 30-inch column of water. The filter shall be examined for wetting at the end of 45 seconds exposure. The use of a water-soluble red dye will facilitate the observation.

4.4.8 Leakage and separation of set. The test shall be applied to those samples subjected to the piercing device durability test (see 4.4.6). The needle, and the cap containing the air inletting bacterial filter shall be removed. The plastic needle adapter of the set shall be fitted into the cavity of the air inlet. The set shall then be immersed in water and subjected to a pneumatic pressure of 8 pounds per square inch gage (psig) for 60 seconds. The pressure shall be applied through the piercing device and the set observed for separation and leakage. These shall be no more than one clinging air bubble.

4.4.8.1 Leakage and drip test. An infusion rate of 40 $\pm$ 5 drops per minute shall be established on the intravenous injection set when set-up as shown in Figure 2. The solution bag shall be squeezed by hand to 100 mm $\pm$ 10 of Hg maintained for 5 seconds, observed for leakage after which the infusion rate shall be measured which shall not vary more than 10 drops per minute from the original infusion rate setting.

4.4.9 Reseal test. The rubber diaphragm shall reseal without a falling drop within 10 seconds after puncture with a nonlubricated 18 gage needle when subjected to an internal pressure (hydrostatic pressure) of 4 p.s.i.g. for 30 seconds.

4.4.10 Clamping device performance test. The assembled set (except for needle) shall be connected, in the normal manner, to a standard solution container as specified in 3.2.2 except, as an alternate, the container may be filled with saline solution of water. The container, with set connected thereto, shall be suspended, in the normal manner, at a sufficient height to permit the entire length of the tubing of the set to hang vertically. The ability of the clamp to accomplish complete shut off, to regulate flow, and with a constant head of water maintain a constant rate of flow, shall be tested with the clamp at a point on the plastic tubing within two inches above the molded rubber part and shall meet the requirement specified in 3.2.5. The tests shall be performed with the tubing hanging vertically at all times. The rate of flow shall be determined either by counting the drops in the drip chamber during a measured time period, or by recording the time required to collect a measured volume of fluid from the needle adapter.

4.4.10.1 Security of roller in clamp body. The roller clamp shall be removed from the set together with a piece of tubing approximately one inch longer than the clamp body. The tubing shall be present in the roller clamp in its normal position with a 1/2 inch length extending from each end of the roller clamp. The clamp (see Figure 3) shall be held vertically, with the large end down, over a supporting gap so that the roller if forced out of the body is free to fall. A polished steel rod is applied to the top of the roller, so that the roller is able to rotate in the course of the test, and the vertical downward force required to separate the roller from the clamp body is measured. As an alternate, any position of the clamp is acceptable providing the clamp body is not distorted and the load measurement is accurate.

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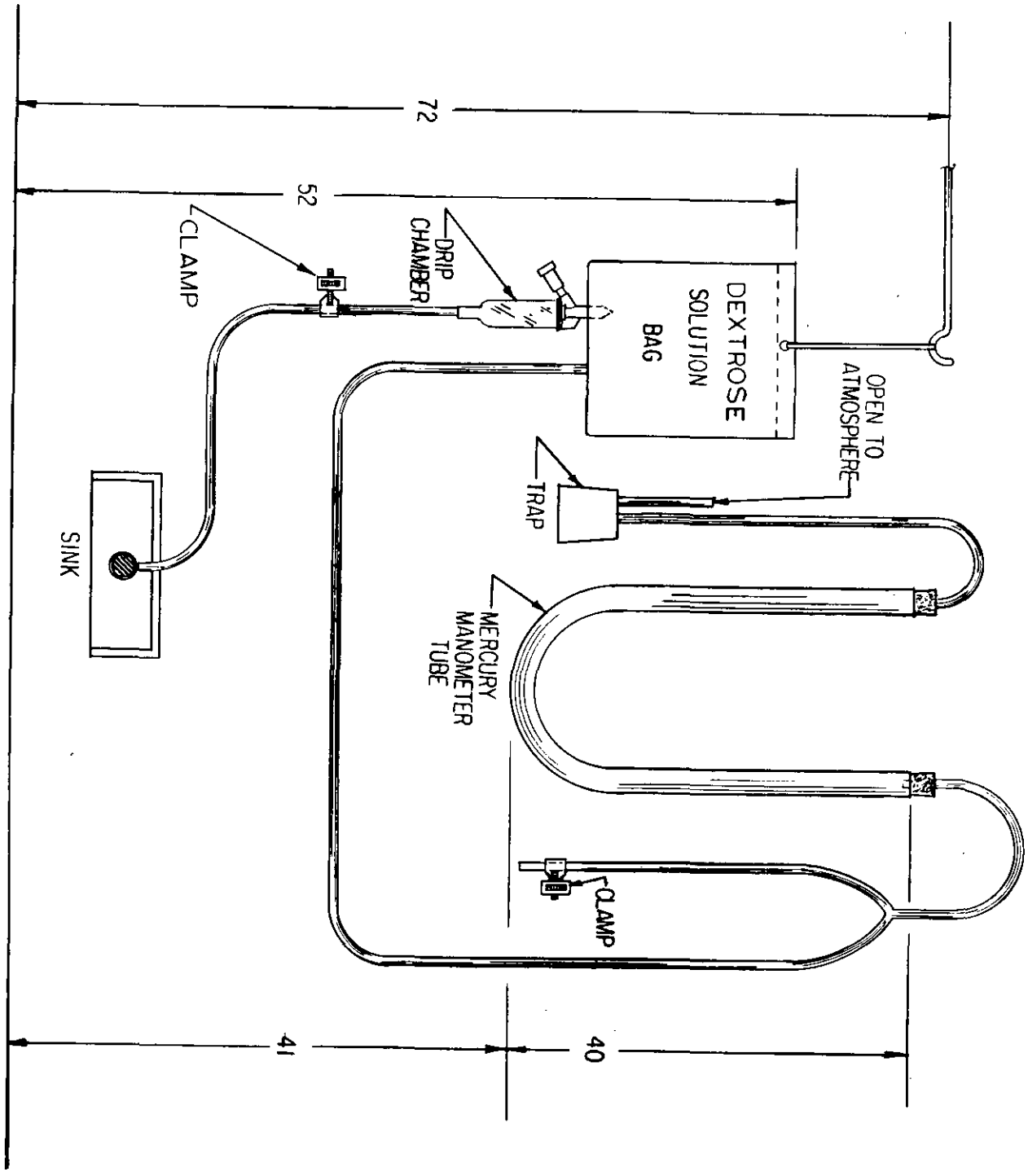
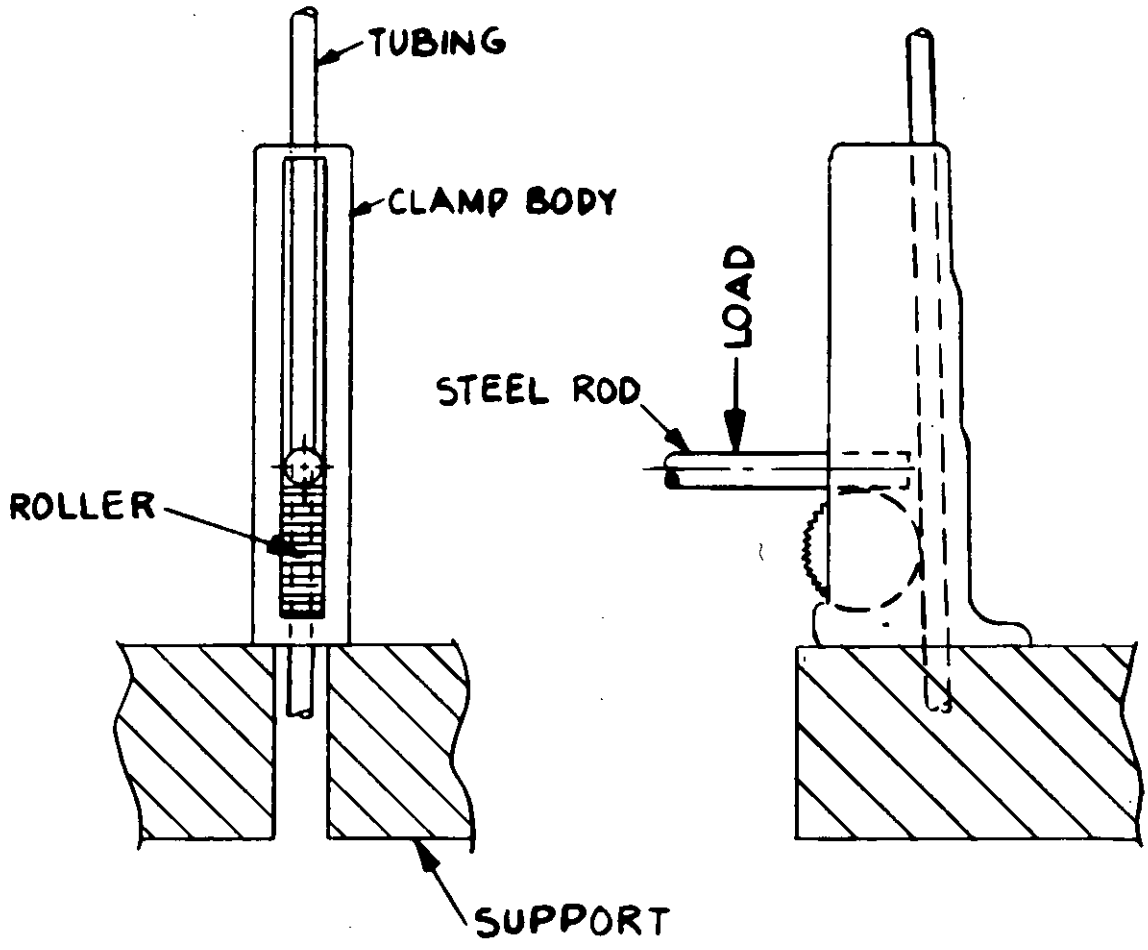


FIGURE 2
LEAKAGE AND DRIP RATE SET-UP

NOTE:
DIMENSIONS ARE IN INCHES AND ARE SUGGESTED ONLY.

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METHOD OF APPLYING LOAD TO ROLLER

FIGURE 3

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4.4.11 Intravenous needle.

4.4.11.1 Diameter and depth of hub cavity. The needle hub cavity shall be tested on the male Luer taper gage in accordance with ANSI/HIMA Standard MD 70.1. (This gage shall be as shown in the ANSI/HIMA Standard MD70.1 with a suitable fluid path to permit performance of the fluid tightness test specified in 4.4.11.2). The fit of the hub on this gage shall be tested in accordance with ANSI/HIMA Standard MD70.1.

4.4.11.2 Fluid tightness. With the needle in horizontal position, the hub shall be attached by means of a plug gage, as shown in Figure 6 of the ANSI/HIMA Standard MD70.1 to a water pressure source having a pressure gage. The water shall be run through the needle so as to fill any hub cavity space and the lumen of the cannula; the lumen shall then be sealed by a piece of solid rubber slipped onto the point, and a pressure of 45 pounds per square inch applied for 10 seconds.

4.4.12 Freedom from pyrogens. Samples of the finished sets shall be tested for freedom from pyrogens, in accordance with the USP. Sampling shall be shown in Table II.

4.4.13 Sterility Test. The intravenous injection sets shall be tested for sterility in accordance with the method specified in the U.S.P. using inoculated product, inoculated simulated product, or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates it from product intended for distribution. The incubation medium for the paper spore strip (carrier) shall be soybean-casein digest. The incubation media for the inoculated product, or inoculated simulated product shall be fluid thioglycollate or alternative thioglycollate at 30° to 35° C for one half of the specimens and soybean-casein digest at 20° to 25° C for the remaining half of the specimens. The minimum incubation time shall be ten days. Sampling shall be as follows:

Medium	Paper Spore Strip	Inoc.Prod. or Inoc.Simulated Prod.
Soybean-casein digest	S-3*	S-3*
Fluid thioglycollate or Alternative thioglycollate		S-3*

*Use a minimum of 20 specimens. Acceptance number is zero.

4.4.13.1 Inoculated product or simulated product For each type of sterilization, the organism used shall be as listed below:

Type of Sterilization	Organism
Steam	B. stearothermophilus
Dry heat	B. stearothermophilus
	B. subtilis
Ethylene oxide	B. subtilis
Gamma radiation	B. pumilus

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The number of viable spores shall be sufficient to provide assurance that the product has been sterilized.

4.4.13.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used, shall be in accordance with MIL-S-36586A, Sterilization Test Strip Set, Bacterial Spore. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of test units shall be as indicated in 4.4.13. Sterility test and 4.4.13.1 Inoculated product or simulated product. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.4.13.1. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the U.S.P. and paragraphs 4.4.13 and 4.4.13.1.

4.4.13.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method on a government contract.

4.4.13.4 Product testing. Testing for sterility of the product after receipt by the government shall be without a biological indicator and in accordance with the U.S.P. The inspection level shall be as stated in paragraph 4.4.13 above. Acceptance number shall be zero.

4.4.14 Safety test. The finished sets shall be tested for safety in accordance with the USP. Sampling shall be as shown in Table II.

4.4.15 Solution flow rate. The finished intravenous injection set shall be connected to a standard solution container as specified in 3.2.2 in the normal manner, the entire assembly inverted and suspended so that there is a 30-inch column of fluid in the set and container and no restriction of flow imposed by the needle or the clamp. The time neccessary for passage of 100 cc. of fluid through the entire set shall be determined. This test procedure shall be repeated after purposely wetting the filter with water as specified in 3.9.

4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. Packaging.

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A.

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5.1.1.1 Individual package. Each intravenous injection set shall have the tubing coiled in such a manner to prevent kinking or taking a permanent set and shall be packaged conforming to MIL-P-116, submethod IC-2.

5.1.1.2 Unit package. Forty-eight intravenous injection sets shall be packaged in a box conforming to PPP-B-636, class domestic. Closure shall be effected with tape or adhesive as specified in the box specification. Metal stitches shall not be used.

5.1.2 Level C. The level C preservation for intravenous injection sets shall conform to MIL-STD-794 requirements for this label.

5.2 Packing. Packing shall be Level A, B, or C as specified (see 6.2).

5.2.1 Level A. Four unit packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the applicable box specification. Fiberboard boxes shall be waterproofed as specified in the box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure shall be as specified in the liner specification.

5.2.2 Level B. Four unit packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. The level C packing shall conform to the MIL-STD-794 requirements for this level.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Levels A and B. Intravenous injection sets, packed as specified in 5.2.1 and 5.2.2, shall be unitized on pallets conforming to MIL-STD-147.

5.2.4.2 Level C. Intravenous injection sets, packed as specified in 5.2.3, shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Individual package. Each package shall be marked in accordance with commercial practice and the requirements of the Federal Food, Drug, and Cosmetic Act. In addition, each unit shall contain instructions as specified in 3.10, and the legend: "Fluid path and end connectors are sterile and pyrogen free." Marking shall include the lot or control number.

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5.3.2 Levels A, B, and C.

5.3.2.1 Unit package. Each unit package shall be marked as specified in MIL-STD-129. Marking shall include the lot or control number and name of the contractor. When the contractor is not the manufacturer, name of manufacturer shall also appear.

5.3.3.2 Each exterior container and unitized load shall be marked as specified in MIL-STD-129. Marking shall include the lot or control number.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Intended use. The intravenous injection set is intended for accurate administration of parenteral solutions from standard glass solution containers.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification is intended to cover the following items appearing in the Federal Supply Catalog:

NSN 6515-00-117-9021	INTRAVENOUS INJECTION SET, Vented, With Flexible Drip Chamber, 19 or 20 Gage Needle, Disposable, 48s (Type I).
NSN 6515-00-118-1341	INTRAVENOUS INJECTION SET, Vented, With Flexible Drip Chamber Without Needle, Pediatric, Disposable, 48s (Type II).

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This specification does not include all types, sizes, or grades as indicated by the title or all type which are commercially available.

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:
Defense Supply Agency-DM

Review Activities:

Army - MD
Navy - MS
Air Force - 03

Project No. 6515-3293

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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